

NIMA: Neural Image Assessment

2018 IEEE TIP

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Model Architecture

- Uses pretrained CNN backbones (e.g., Inception-v2, MobileNet, VGG16)
- Output: a **probability distribution over scores 1-10**
- Softmax used for final layer
- Loss: **Earth Mover's Distance (EMD)** between predicted and ground truth distributions

What is Earth Mover's Distance (EMD)?

- EMD measures **how much "effort" is needed to turn one distribution into another**
- Inspired by a transportation analogy:

"How much work is needed to move dirt from one pile (prediction) to match another (ground truth)?"

- ✓ Takes **ordering and distance between score bins** into account
- ✓ Better than cross-entropy when labels have ordinal meaning (e.g., image scores)

EMD Formula in NIMA

Let:

- $p = [p_1, \dots, p_N]$: predicted distribution
- $q = [q_1, \dots, q_N]$: ground-truth distribution
- $N = 10$: for scores from 1 to 10

$$EMD(p, q) = \left(\frac{1}{N} \sum_{k=1}^N \left(\sum_{i=1}^k (p_i - q_i) \right)^2 \right)^{1/2}$$

Interpretation:

- $\sum_{i=1}^k (p_i - q_i)$: cumulative error up to score k
- Squares penalize large shifts in mass
- Final root returns to original scale

Visual Intuition

Imagine this:

- Ground-truth: mostly scores 4-6
- Prediction: shifted to 7-9

EMD will accumulate the **magnitude and direction of misalignment**, punishing distant shifts more than nearby ones.

- Cross-entropy: "Did you guess right?"
- EMD: "How far off were you and in which direction?"

Why EMD Matters for NIMA

Property	EMD	Cross-Entropy	MSE
Captures ordinal nature	✓	✗	✗
Models full distribution	✓	✓	✗
Sensitive to distribution shift	✓	✗	✓
Works well for subjective tasks	✓	✗	✗

NIMA uses EMD to reflect the **subjective and graded nature of image quality**.

Training Datasets

Dataset	Purpose	Notes
AVA	Aesthetic scores	~200 ratings per image, scores from 1-10
TID2013	Quality assessment	Includes distortion types and MOS (Mean Opinion Score)

Results & Applications

- **Strong correlation** with human judgments
- Predicts not just mean score but **score distribution**
- Useful for:
 - Ranking/filtering photos
 - Evaluating image enhancement quality
 - Reward function for generative models

Summary

Aspect	Description
Goal	Predict subjective image quality or aesthetics
Model	CNN + Softmax output
Output	Distribution over scores 1-10
Loss	Earth Mover's Distance (EMD)
Datasets	AVA (aesthetics), TID2013 (quality)
Advantages	Captures human subjectivity and score uncertainty

Extensions & Applications

- **Enhancement reward:** Used in EnhanceNet, Deep Photo Enhancer
- Can be extended to **Video Quality Assessment (VQA)**
- Useful for training **generative or enhancement models** with perceptual goals

Q&A

Thanks for listening!